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Article

Nationwide Study of the Association between Temperature Variability and Sleep Apnea, and the Modifying Role of Air Pollution, in a Moderate-to-Severe Risk OSA Population

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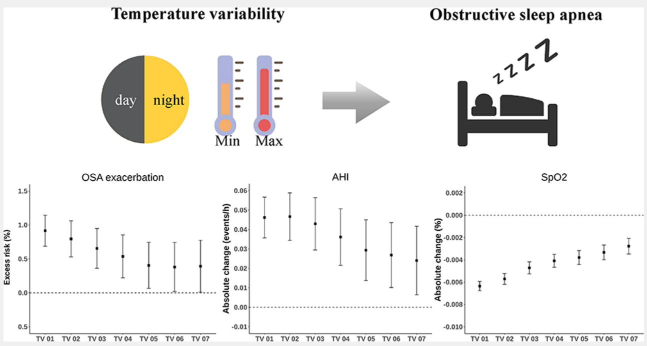
ABSTRACT: Obstructive sleep apnea (OSA) is a severe sleep disorder that is associated with various health complications. Despite previous research investigating the impact of ambient temperature on the severity of OSA, the potential influence of temperature variability (TV) remains underexplored. We used a nationwide data set among wearable smart device users to assess the association between TV and OSA severity in a population at moderate-to-severe risk of OSA. A case time series statistical method was used to assess the associations between TV and the risk of OSA exacerbation, apnea-hypopnea index (AHI), and mean oxygen saturation (SpO_2). Additionally, we explored the interaction of TV and air pollution with OSA severity. A total of 52,752 participants with 6,349,414 days of monitoring were finally included in the study. A 1 °C increase in TV_{01} was associated with an increase of 0.92% (95%CI: 0.69, 1.15%) in excess risk of OSA exacerbation, increases of 0.046 events/h (95%CI: 0.036, 0.057) in AHI, and a decrease of 0.0064% (95%CI: 0.0059, 0.0068%) in SpO_2. We observed a significant synergistic interaction of TV and air pollution on the OSA severity. This study highlights that TV may exacerbate the severity of OSA, and air pollution can synergistically interact with TV to impact the OSA severity.



KEYWORDS: *temperature variability, obstructive sleep apnea, apnea-hypopnea index, oxygen saturation, case time series*

■ INTRODUCTION

Sleep disorders are prevalent in the general population, affecting up to one-third of adults.[1] Obstructive sleep apnea (OSA), a particularly severe sleep disorder, has been identified as an independent risk factor for cardiovascular and neurocognitive diseases.[2] Despite affecting nearly one billion people globally, OSA remains insufficiently recognized and treated due to the complexity of its diagnosis (i.e., polysomnography, PSG),[3] which requires specialized equipment and technical personnel and potentially disrupts lifestyles during diagnosis. Therefore, identifying risk factors for OSA exacerbation is crucial to reducing the disease burden and its associated consequences. Existing studies indicated that, beyond prevalent physiological factors[4] and lifestyle habits,[5] OSA may also be associated with environmental factors, including air pollution and temperature.

Environmental factors affecting sleep have gained widespread attention.[6] Previous studies showed that rising temperatures can increase the severity of OSA, with short-term temperature spikes aggravating the OSA condition.[7−10] However, current epidemiological research often overlooks the impact of temperature variability (TV). TV is an independent health risk factor[11−13] that offers advantages in assessing temperature adaptability,[14,15] which introduces another temporal dimension. It effectively captures long-term health trends, accounts for temperature acclimatization, and reveals regional variations.[16] However, the impact of TV on the severity of OSA remains unexplored, which could have important implications for managing OSA risk populations in terms of environmental risks. Recent epidemiological studies also suggest that extreme temperatures and air pollution can synergistically pose health hazards.[17−19] For instance, research has shown that the combined exposure to high temperatures and elevated levels of air pollutants such as fine particulate matter ($PM_{2.5}$) presents a greater risk to respiratory health than the sum of their individual effects.[20] This synergistic effect underscores the importance of considering the interplay

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between temperature and air pollution when assessing their impact on health. However, whether there is a synergistic effect between TV and air pollution to elevate the risk of OSA exacerbation remains unknown. Additionally, moderate to severe OSA poses major health risks and increases comorbidity risks, severely impacting the quality of life. Effective treatment results in significant health improvements, highlighting the need for strategies targeting this high-risk group to address urgent clinical needs and societal impacts.

To address these gaps, we performed a population-based longitudinal study using data from more than 6.3 million days of sleep records in 328 Chinese cities from 2019 to 2022. This study aimed to comprehensively assess the association between TV exposure and OSA severity and to quantitatively assess the interaction between TV and air pollutants on OSA severity.

■ METHODS

Study Design

This study used data from the premobile atrial fibrillation app (mAFA) II registry, and all participants participated in the study voluntarily and signed an electronic informed consent form. All participants were monitored with photopolymorphography (PPG)-based Huawei smart devices (Huawei Technologies Co.). In this study, we included 53,552 participants aged ≥18 years, identified as having moderate-to-severe OSA risk from December 16, 2019, to October 15, 2022. Participants were identified as moderate-to-severe OSA risk and included in analysis if there were 80% or more nights with AHI ≥ 15 in any two-week period with a cumulative occurrence of five or more sleep monitoring during the study period. Individuals with mild OSA risk were excluded due to their low frequency of exacerbations, as they are not considered a high-risk group for clinical intervention. Data authorization received approval from the Central Medical Ethics Committee of the Chinese People's Liberation Army General Hospital (S2017-105-02). The data used in this analysis were completely anonymous. Participants were required to complete a questionnaire gathering essential demographic information, including age, gender, height, weight, residential address, and the history of diseases.

We applied a novel case time series design to analyze the longitudinal data, which combines elements of traditional time series methods with self-matched methods like the case-crossover approach.[21] This approach allows for flexible control of time-varying confounders within a longitudinal structure and automatically addresses time-invariant confounders through a self-controlled structure.

Sleep Monitoring Data

We assessed the severity of OSA using three indicators: OSA exacerbation, AHI, and SpO_2, which were calculated based on sleep monitoring data. All sleep monitoring data were obtained using the Huawei smartwatch GT2, capable of capturing PPG signals through an optical heart rate sensor and wrist pulse oximeter.[22] The data collected was captured in 1 min intervals. The raw signals underwent processing by the company's backend systems, and the data were summarized as nightly records for each user. The smartwatch was developed by the company and empowered by PPG signals and machine learning algorithms, which generated a PPG-derived respiration (PDR) waveform and mean oxygen saturation (SpO_2) feature. Hypopnea and apnea events were defined as reductions in PDR by 30 and 90%, respectively.[23,24] The device utilized infrared and red-light PPG signals to obtain oxygen saturation levels by analyzing the differential absorption coefficients between hemoglobin and deoxyhemoglobin,[25,26] validated for robust reliability and accuracy. Additionally, the smartwatch is also equipped with an accelerometer (ACC) sensor to detect motion parameters, including duration/frequency, amplitude/intensity, and acceleration/speed.[27] The accelerometer signals can be used to distinguish the status of wakefulness or sleep, as well as the sleep onset and cessation time points for night-time sleep.[28] The apnea-hypopnea index (AHI) was calculated as the sum of hypopnea and apnea events divided by the total sleep time (events/h) per night. SpO_2 was identified as the mean oxygen saturation level (%) observed throughout the entire duration of sleep, which currently serves as a guiding indicator for OSA severity.[29] Among these populations with moderate-to-severe OSA risk, OSA exacerbation was categorized as a binary variable for each night, indicating whether the concomitant AHI for that day was ≥5 for individuals.[10,30−33]

The sleep monitoring data were not required to be continuous and were not limited to a specific month, season, or year. Meanwhile, daily sleep monitoring lasting 4 to 12 h was considered eligible for inclusion to ensure comparability of sleep periods. When compared with polysomnography (PSG), the gold standard for diagnosing OSA, the PPG-based technique exhibited high sensitivity and specificity in screening for OSA.[34−37] We conducted PSG on a subset of participants to ensure the accuracy of our screening method. A total of 1782 participants screened for moderate to severe OSA risk by smartwatches received PSG testing in hospitals, and 93.2% were ultimately diagnosed with OSA, indicating the high accuracy of our outcome ascertainment. Furthermore, it is worth noting that Huawei smart devices for OSA screening have received official validation and recognition from the China Food and Drug Administration.

Environmental Data

Meteorological data (i.e., temperature and relative humidity) were obtained from ERA5-Land data with 0.1° × 0.1° spatial and hourly temporal resolutions. To account for air pollutants, we assessed $PM_{2.5}$, coarse particulate matter ($PM_{2.5-10}$ calculated by PM_{10} minus $PM_{2.5}$), sulfur dioxide (SO_2), nitrogen dioxide (NO_2), and carbon monoxide (CO) from China's National Urban Air Quality Real-time Publishing Platform (https://air.cnemc.cn:18007/). The exposure levels of environmental factors for each participant were determined by assigning data from the nearest monitoring station or grid cell based on the center of the residential district for privacy reasons. Participants residing more than 50 km from the nearest air pollution monitoring station were excluded to reduce exposure measurement errors. To mitigate the influence of outliers, we excluded data points outside three standard deviations from the mean in meteorological data (i.e., values beyond the 0.27 and 99.73% intervals of a normal distribution).

For each participant, we defined T_{max} − lag0 and T_{min} − lag0 as the maximum and minimum hour-averaged temperatures within 24 h before waking time. We calculated the moving maximum and minimum temperature exposures on multiple lag days. TV was determined as the standard deviation (SD) of daily T_{min} and T_{max} over several days. For instance, TV at lag 0−1 (TV_{01}) was calculated as follows: TV_{01} = SD (T_{min} − lag0, T_{max} − lag0, T_{min} − lag1, T_{max} − lag1).[38] We evaluated the association between TV and OSA exacerbation, AHI, and SpO_2 for multiple lags from lag 0−1 to lag 0−7.

Statistical Analyses

The case time series design is a novel self-matched method proposed by Gasparrini[21] for the analysis of transient changes in risk of acute effects associated with time-varying exposures. It splits the follow-up period into a daily time series for each case, yielding a set of multiple case-specific time series. Fixed-effects regression models were used to estimate the association between TV and the risk of OSA exacerbation, AHI, and SpO_2. In the main models, we incorporated the subject/year/month strata intercept to control for the individual-level variations, as well as the yearly and monthly variations.

A case time series model can be written in a regression form by defining the expectation of a given health outcome y_{it} for case i at time t in relation to a series of predictor terms.[21] Thus, y_{it} sequentially represents a binary indicator for OSA exacerbation, continuous values of AHI (events/h), and SpO_2 (%). Algebraically, the model can be written as

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$$g[E(y_{it})] = TV_{it} + \sum_{j=1}^{J} s_j(t) + \sum_{p=1}^{P} h_p(z_{ipt}) + \xi_{i(k)}$$

(1) Where g is a logarithm link function, the binomial family was used for binary OSA exacerbation, and the Gaussian family was used for the other two continuous results (AHI and SpO_2). (2) The TV_{it} represents the exposure values of TV for case i at time t. (3) The term(s) S_j represent functions expressed at different time scales to model temporal variations in risk associated with underlying trends or seasonality, among others. We included a natural spline of calendar day with 7 degrees of freedom (df) per year to adjust for the long-term trends, an indicator variable for the "day of the week," and a factor variable for public holidays, represented via $s(t)$. (4) Other measurable time-varying confounders z_p can be modeled through functions h_p. Further, we control for mean temperature with 3 df, relative humidity with 6 df, and mean wind speed with 3 df, represented via $h(z_{it})$. The two sets of terms, S_j and h_p, ensure strict control of temporal variation in risks over multiple time axes. (5) Finally, intercepts $\xi_{i(k)}$ are defined for each stratum k corresponding to month-in-year subsets of the study period. For OSA exacerbations, the estimated risk was transformed into the percentage change in excess risk (ER) associated with each 1 °C increase in TV, which was calculated using the following formula: $(e^{\beta\times1} - 1) \times 100\%$.[39] For AHI and SpO_2, the absolute changes were presented as β. The β is the regression coefficient from the fixed-effect regression models for corresponding outcomes. To examine potential nonlinear relationships between TV and the severity of OSA, the exposure−response curve was modeled by natural cubic spline with 3 knots, showing variation in OSA severity across the range of TV.

Stratified analyses explored effect modifications by age (<45 vs ≥45 years), BMI (<28 vs ≥28 kg/m^2), history of hypertension (yes/no), history of diabetes (yes/no), season (warm season: April to September; cool season: October to March),[40] heating supply (yes/no, based on official Chinese Web site for the heating supply dates for each city), the level of monitoring days (high/low) and region (north/south)[41] regarding the association between TV and OSA severity. Between-group differences of health effects were assessed using Z-statistics as follows:

$$Z = \frac{\beta_1 - \beta_2}{\sqrt{SE_1^2 + SE_2^2}}$$

where β_1 and β_2 were the OR estimates of two subgroups, and SE_1 and SE_2 were their respective standard errors.[34,42] Then, the two-sided P value corresponding to the Z-statistic was derived from the standard normal distribution. This P value reflects the probability of observing the subgroup differences under the null hypothesis of no heterogeneity.

Several sensitivity analyses were conducted to ensure the robustness of our findings. First, we excluded any data (2.9%) where the bedtime was earlier than 19:00 or later than 03:00, and wake-up time was earlier than 03:00 or later than 11:00 to address potential heterogeneity in daytime and nighttime sleep.[43] Additionally, we performed a sensitivity analysis by redefining a day from midnight to midnight rather than the 24 h period before each individual's wake-up time. TV was recalculated using maximum, minimum, and mean temperatures from midnight to midnight. Third, we investigated the association between TV and OSA severity by additionally adjusting for the length of sleep in the main models to account for the potential confounding. Lastly, we examined the association between TV and OSA severity by adjusting for five pollutants ($PM_{2.5}$, $PM_{2.5-10}$, SO_2, NO_2, and CO) in the main models to account for the potential confounding of air pollution.

Additionally, we categorized the exposure levels of each air pollutant into binary variables based on the median and incorporated them into the model to evaluate the p-values of multiplicative interaction terms between TV as a continuous variable and air pollution.

The above statistical analyses were performed in R software (version 3.6.3). Regression analyses utilized the survival and gnm packages, and plotting employed the ggplot2 packages. All statistical tests are two-sided, and p-values below 0.05 indicate statistical significance.

■ RESULTS

Descriptive Statistics

During the study period, 53,552 individuals at moderate to severe OSA risk were included. We excluded those with missing residential address information, individuals living over 50 km from the nearest monitoring station or air quality monitoring stations (n = 476), as well as those with fewer than 7 days of eligible monitoring (n = 324). Consequently, 52,752 participants with 6,349,414 days of eligible monitoring were included in the analysis (Figure S1). The average of eligible monitoring was 120 days for each subject. The average age of the participants was 45.4 years (standard deviation, SD = 10.9), and 95.5% were men. A total of 30.5% of participants had a BMI of ≥28 kg/m^2, 28.7% had a history of hypertension, and 5.6% had a history of diabetes (Table 1). Most (97.1%) of the sleep monitoring data were classified as nighttime sleep.

Table 1. Descriptive Statistics on Study Participants[a]

variables	overall (N = 52,752)
age, mean (SD), year	45.4 (10.9)
Sex, %	
female	2348 (4.5%)
male	50,404 (95.5%)
Body Mass Index, No. (%)	
<28 kg/m^2	29,759 (56.3%)
≥28 kg/m^2	16,081 (30.5%)
missing	6912 (13.1%)
Hypertension, %	
no	22,535 (42.7%)
yes	15,123 (28.7%)
missing	15,094 (28.6%)
Diabetes, %	
no	35,299 (66.9%)
yes	2967 (5.6%)
missing	14,486 (27.4%)

[a]Abbreviations: SD, standard deviation.

As summarized in Table 2, throughout the all-sleep periods, we identified a total of 2,930,405 days of OSA exacerbation, constituting 46.2% of all eligible monitoring days. The median AHI levels were 0.90 events/h in person-days without OSA exacerbation and 30.68 events/h in person-days with OSA exacerbation. The averages of SpO_2 were 96.50 and 96.30% in these respective categories. The average TV_{01} was 5.16 and 5.21 °C for person-days without and with OSA exacerbation (Table S1). The mean concentrations of $PM_{2.5}$, $PM_{2.5-10}$, SO_2, NO_2, and CO at lag 0−1 in person-days with OSA exacerbation were slightly higher (32.51 $\mu g/m^3$, 26.69 $\mu g/m^3$, 7.92 $\mu g/m^3$, 29.08 $\mu g/m^3$, and 0.69 mg/m^3, respectively) compared to those without OSA exacerbation.

Regression Results

Figure 1 presents the percentage changes in OSA exacerbation, absolute changes of AHI and SpO_2 associated with a 1 °C increase in TV across different lagged exposures, ranging from lag 0−1 to lag 0−7. Our study revealed an elevated OSA risk,

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Table 2. Summary Descriptive Statistics on OSA-Related Indicators, Weather Conditions, Air Pollution, and Temperature Variability[a]

variables (mean, SD)	person-days without OSA exacerbation (N = 3,419,009)	person-days with OSA exacerbation (N = 2,930,405)
OSA-Related Indicators		
AHI (events/h)[b]	0.90 (0.41)	30.68 (11.39)
SpO_2 (%)	96.50 (0.75)	96.30 (0.73)
Weather		
TV_{01}	5.16 (1.98)	5.21 (1.98)
TV_{02}	5.03 (1.82)	5.08 (1.81)
TV_{03}	5.00 (1.74)	5.05 (1.73)
TV_{04}	5.00 (1.69)	5.06 (1.68)
TV_{05}	5.01 (1.66)	5.07 (1.65)
TV_{06}	5.03 (1.64)	5.09 (1.63)
TV_{07}	5.05 (1.62)	5.11 (1.61)
temperature (°C)[c]	16.70 (9.71)	16.34 (9.60)
humidity (%)[c]	65.68 (18.16)	65.64 (18.08)
Pollutants[c]		
$PM_{2.5}$ (μg/m^3)	31.37 (23.45)	32.51 (24.01)
$PM_{2.5-10}$ (μg/m^3)	25.70 (23.71)	26.69 (24.86)
SO_2 (μg/m^3)	7.72 (4.83)	7.92 (4.90)
NO_2 (μg/m3)	28.60 (15.80)	29.08 (15.95)
CO (mg/m^3)	0.69 (0.27)	0.69 (0.27)

[a]Abbreviations: OSA, obstructive sleep apnea; AHI, apnea-hypopnea index; SpO_2, mean oxygen saturation; TV_{01}, temperature variability at lag 0−1, and so on; $PM_{2.5-10}$, coarse particulate matter; $PM_{2.5}$, particulate matter with an aerodynamic diameter less than or equal to 2.5 μm; NO_2, nitrogen dioxide; SO_2, sulfur dioxide; CO, carbon monoxide; SD, standard deviation; IQR, interquartile range. [b]Median (IQR). [c]Weather conditions and air pollution levels in 0−1 moving days before waking time.

an increase in AHI, and a decrease in SpO_2 with higher TV. The effect estimates varied for lags of TV, which was the strongest at TV_{01} and decreased from lag0−1 to lag0−7 of TV. A 1 °C increase in TV_{01} was associated with an increase of 0.92% (95%CI: 0.69, 1.15%) in ER of OSA exacerbation, increases of 0.046 events/h (95%CI: 0.036, 0.057) in AHI, and a decrease of 0.0064% (95%CI: 0.0059, 0.0068%) in SpO_2 (Table S2)

The exposure−response curves for associations between TV_{01} and OSA exacerbation, AHI, and SpO_2 are shown in Figure 2. Overall, the curves illustrated a monotonically increasing ER of OSA exacerbation as well as AHI with rising TV_{01} exposures, and a nearly linear association between increased TV_{01} and decreased SpO_2.

The stratified analysis reveals significant differences across various demographics and environmental conditions, including age, sex, BMI, prevalence of hypertension and diabetes, season, heating supply, region, and monitoring levels (Table S3). Notably, the association between TV and the severity of OSA was more pronounced in individuals aged 45 and older, females, and in southern regions, as well as with a high level of monitoring, showing significant between-group differences ($p < 0.05$).

Table S4 summarizes the results of all of the sensitivity analyses. The associations of TV_{01} with all three OSA severity indicators remained robust after daytime sleep monitoring data. Furthermore, in the supplementary analysis to investigate the association between a shorter exposure on the same nights and OSA severity, we redefined a day from midnight to midnight and recalculated TV and its association with OSA severity indicators based on this new definition; the results stayed statistically significant. The associations of TV_{01} with all three OSA severity indicators remained robust after additionally adjusting for the length of sleep. Lastly, the association was slightly reduced but remained robust after additionally adjusting for all five air pollutants, respectively.

Table 3 summarizes the interactive effects of TV and air pollution on the OSA exacerbation. Analyses for the inclusion of interaction terms showed that there was effect modification of $PM_{2.5}$ and SO_2 for TV exposure on all OSA-related indicators, $PM_{2.5-10}$ for TV exposure on AHI and SpO_2, and NO_2 and CO for TV exposure on OSA exacerbation and AHI.

■ DISCUSSION

In this population-based longitudinal study spanning three years, covering 328 major Chinese cities with 52,752 participants and over 6.3 million monitoring days, we found that higher temperature fluctuations were associated with an increased risk of OSA severity, and air pollution can synergistically interact with TV to impact OSA severity. The association between TV and OSA severity was stronger at lag 0−1 than at longer lag days. Furthermore, this association was

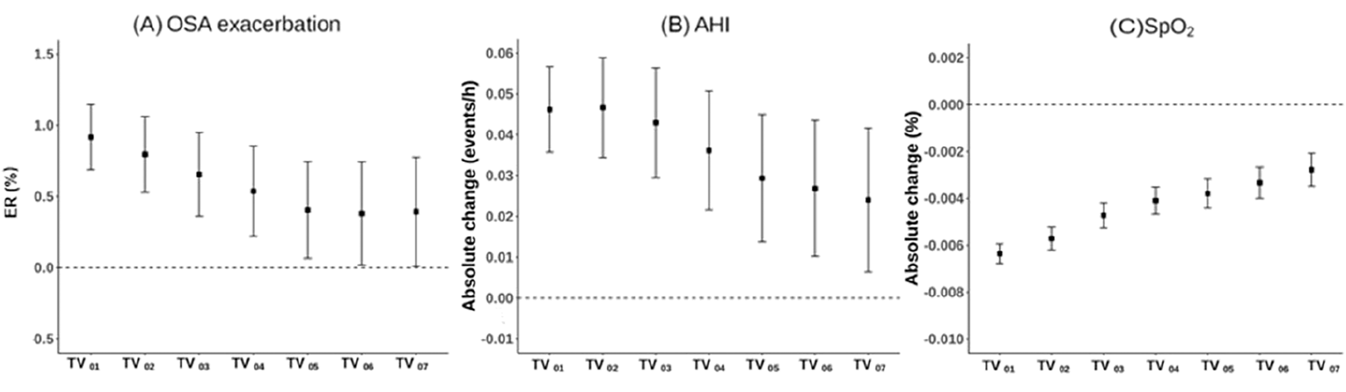

Figure 1. ER (and 95% CI) of OSA exacerbation (A), absolute changes (and 95% CI) in AHI (B), and SpO_2 (C) per 1 °C TV increase over different lag periods in the main analyses. Abbreviations: OSA, obstructive sleep apnea; AHI, apnea-hypopnea index; SpO_2, mean oxygen saturation; TV, temperature variability; ER, excess risk.

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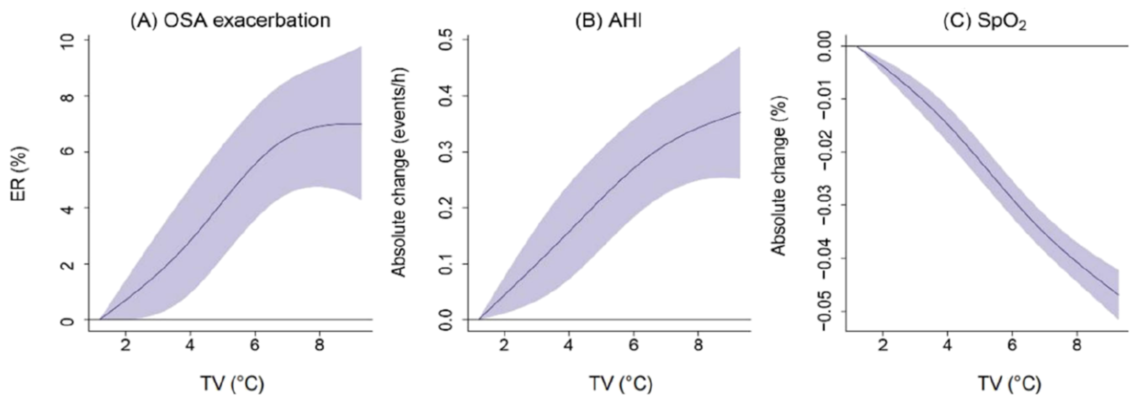

Figure 2. Exposure−response curves for the association of TV with OSA exacerbation (A), AHI (B), and SpO_2 (C) at lag 0−1. Solid lines= percentage change of ER in OSA exacerbation or absolute change of AHI and SpO_2; shaded areas = 95% confidence intervals. Abbreviations: OSA, obstructive sleep apnea; AHI, apnea-hypopnea index; SpO_2, mean oxygen saturation; TV, temperature variability; ER, excess risk.

Table 3. Coefficients and *p*-Values for the Interaction Terms of TV_{01} and Air Pollutants in the Association between TV_{01} on OSA-Related Indicators[a]

air pollutant		OSA exacerbation	AHI	SpO_2
$PM_{2.5}$	β	0.005	0.029	−0.00217
	p	0.001	<0.001	<0.001
$PM_{2.5-10}$	β	0.003	0.016	−0.00193
	p	0.124	0.037	<0.001
SO_2	β	0.006	0.029	−0.00189
	p	<0.001	<0.001	<0.001
NO_2	β	0.008	0.037	−0.00039
	p	<0.001	<0.001	0.147
CO	β	0.003	0.013	−0.00032
	p	0.027	0.056	0.222

[a]Abbreviations: OSA, obstructive sleep apnea; AHI, apnea-hypopnea index; SpO_2, mean oxygen saturation; CI, confidence intervals. $PM_{2.5-10}$, coarse particulate matter; $PM_{2.5}$, particulate matter with an aerodynamic diameter less than or equal to 2.5 μm; NO_2, nitrogen dioxide; SO_2, sulfur dioxide; CO, carbon monoxide; TV_{01}, temperature variability at lag 0−1.

more pronounced in older individuals, females, southern regions, and those with high-level monitoring.

To the best of our knowledge, this is the first nationwide longitudinal study to investigate the association between TV and OSA severity. Previous studies provided a foundation for epidemiological research. First, prior studies indicate that temperature can adversely affect sleep quality.[44−46] For instance, research shows that high temperatures can increase the probability of reporting poor sleep experiences in China,[47,48] leading to increased sleep fragmentation, reduced deep sleep,[49] and more frequent microawakenings. Consequently, OSA patients exposed to temperature variations may experience a decline in their sleep quality, which could worsen the severity of their condition over time. Second, most studies suggest a link between temperature and AHI on the day of screening.[50,51] Our previous research found that high temperatures immediately worsen the risk of OSA.[10] A comparative study of 7523 in-laboratory patients found that AHI followed a circannual pattern, with more events recorded in winter than in other seasons.[52] However, the relationship between ambient temperature and OSA severity remains inconsistent across studies.[6] A cross-sectional study of 5204 adults at the Taipei Sleep Center found that increased temperature was associated with decreased AHI.[53] In contrast, a sectional observational study with 63 participants found no statistically significant relationship between AHI and temperature.[54] The human body's thermoregulatory system may struggle to adapt to sudden temperature changes, and this challenge is even more pronounced in OSA patients, which may be one of the reasons why temperature changes can influence AHI. Finally, some epidemiological studies have found associations between temperature changes and respiratory diseases.[55,56] A prospective Chinese elderly cohort study found an increased hazard ratio of respiratory diseases associated with greater TV.[57] Given that OSA is a common respiratory disorder, it is plausible that it may also be affected by temperature change.

Understanding how ambient temperature over the entire exposure period[58,59] reflects current environmental conditions is crucial for realistically assessing the association between temperature fluctuations and OSA. However, to our knowledge, no prior studies have explored lag patterns in the association between TV and sleep apnea, which adds a valuable insight to the importance of early warning. We calculated the TV values using different exposure windows to better understand the lag effects. Our findings suggest that abrupt changes in temperature over shorter periods significantly impact the severity of OSA. Given the serious health implications of OSA, including increased risks of cardiovascular diseases, metabolic disorders, and reduced quality of life,[60,61] this research takes on added significance. This finding highlights the importance of focusing on short-term, intense temperature changes as potentially more detrimental to those at risk for OSA, informing future strategies for the prevention and management of OSA exacerbations.

Significant temperature fluctuations can plausibly impact the severity of OSA at physiological, immunological, and behavioral levels. First, from a physiological perspective, abrupt temperature changes can disrupt the central respiratory control system, potentially leading to irregular breathing patterns, which may induce decreased oxygen saturation and trigger OSA.[62] Temperature changes can also alter the viscosity of respiratory tract secretions,[63] making them thicker and more likely to obstruct the airway, which can lead to airway obstruction. Second, rapid temperature variations can place stress on the immunology of the body, reducing the ability of the immune system to resist infections.[64] This can potentially result in upper respiratory tract infections, worsening nasal congestion, and airway inflammation, thereby exacerbating

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hypopnea or apnea events. Additionally, temperature fluctuations can influence the production of inflammatory cytokines.[65] Studies have shown that temperature changes can trigger the release of pro-inflammatory cytokines, which promote airway inflammation and contribute to the pathogenesis of OSA.[66] Lastly, behavioral factors play a significant role, as individuals in environments with substantial temperature fluctuations may alter sleep habits and bedtime practices,[67] disrupting normal sleep patterns, adversely impacting sleep quality, and potentially exacerbating OSA.

As the climate crisis advances, air pollution has been identified as a factor that can aggravate the health impacts of meteorological elements.[68−70] TV should be considered as another dimension of temperature, and exploring its health effects should also account for pollutant synergies. Numerous studies have highlighted air pollutants as significant risk factors for OSA.[71,72] However, epidemiological studies have rarely addressed the combined effects of air pollution and TV on the OSA risk. Our findings suggest that when examining the health impacts of temperature changes, it is crucial to also account for the interactions with air pollution, as these factors may together amplify health risks. To the best of our knowledge, this study is the first systematic investigation into the interactive effects of TV and air pollution on OSA severity. Our findings align with prior research on TV and air pollution on respiratory-system-related health outcomes. Notably, a time series analysis showed that airborne particle exposure was linked to heightened risks of the daily temperature range on respiratory infections in Argentina.[73] Previous studies have demonstrated biologically plausible mechanisms, indicating that air pollution induces epithelial injury and inflammatory cytokine production,[74−76] potentially exacerbating the health effects of TV.

Subgroup analyses indicate that the participants aged 45 and older, females, and in southern regions, as well as those with a high level of monitoring, show are more pronounced association between TV and the severity of OSA. Elderly individuals may experience diminished thermal regulation and reduced perspiration rates,[77,78] thus weakening their adaptability to environmental factors. Advancing age brings about a decline in muscle and respiratory function,[79] as well as a decrease in sleep quality. Temperature variations may exacerbate these issues, making the OSA more likely to occur. Sex differences may be related to individual body structure,[80] hormone levels, and immune function.[81] This, coupled with temperature fluctuations, further elevates the risk of OSA. Given the predominantly male population in our study, future research should further explore the modifying effects of sex on the association between TV and OSA. Additionally, we observed notable heterogeneity across regions, with stronger effects in the southern areas. These differences may be attributed to geographic factors,[82] such as latitude, and demographic characteristics like the proportion of elderly residents. Additionally, lifestyle habits,[83] including the intensity of outdoor activities and the usage of air conditioning, alongside economic variables, such as disposable income per capita, could also play a significant role. Therefore, targeted interventions and monitoring of these high-risk groups are of paramount importance.

Our study provides insights that may contribute to the understanding of the association between the environment and the severity of OSA. First, the study underscores the hazardous impact of large TVs and therefore indicates a novel risk factor for managing OSA in high-risk populations in the future. Our findings emphasize the importance of considering TV in relation to OSA severity. It underscores the potential need for future research to explore how interventions, such as indoor air conditioning and heating systems, could impact health outcomes associated with temperature fluctuations. Second, the synergistic effect of TV and air pollution on OSA severity was found in this study, highlighting the need to be wary of high pollution in weather with large temperature changes. For example, an integrated risk warning system that takes into account both temperature and air pollution levels may serve the OSA population more effectively, reducing the adverse health effects of coexposure to environmental risk factors.

However, we also acknowledge several limitations. First, consistent with most prior epidemiological studies on temperature, we relied on ambient meteorological data and air pollution data, instead of personal measurements. Due to data limitations, we were unable to take indoor nighttime temperature into consideration, which might lead to potential exposure measurement errors. However, the case time series approach provides flexible control over time-varying confounders and automatically addresses time-invariant confounders. Furthermore, we had adjusted for long-term time trends and seasons in the models to address any unmeasured time-varying confounding (e.g., air conditioning use). Second, we did not consider factors such as noise and light exposure. However, these factors are unlikely to introduce significant bias into our results because they tend to relate to personal behavior, may remain relatively stable over days, weeks, and even months for the same participants, and can be automatically controlled in the self-control design of CTS. Third, including subjects within a 50 km radius of monitoring stations may introduce exposure misclassification for spatially varying pollutants. However, ambient monitoring data measured based on nearby fixed-site monitoring have been widely used in epidemiological studies investigating the short-term effects of air pollution, and such surrogates have been reported to tend to bias the effect estimates null.[84−86] Fourth, we did not collect data on OSA treatment, such as CPAP therapy. However, considering that the use of CPAP and other standard treatments among OSA patients in China is still relatively low, we postulate that this limitation is unlikely to significantly affect the overall conclusions of our study. Fifth, temperature deviations may affect the accuracy of the OSA screening. Due to commercial confidentiality, verification of Huawei smartwatches based on PPG has not been widely discussed. This limitation should be carefully considered when assessing the applicability of our results to clinical or broader scenarios. Future research should explore Huawei smartwatch performance under varying environmental conditions to gain a more comprehensive understanding of its robustness. However, methodologies for OSA monitoring based on PPG signals have been validated.[34−37] We used the gold standard for OSA screening in follow-up to our study and achieved an accuracy of 93.2% with our method. Last, our study focused on individuals at risk for moderate to severe OSA, which somewhat limits the generalizability of our findings into the general population. However, given that this population is a key target for clinical intervention, our results can still provide valuable clinical insights. Well, with only 4.5% of all participants being female, this may limit the generalizability of our findings. Therefore, further studies should investigate

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the effect of sex modification. In summary, further research is warranted to validate the findings for broader insights.

■ CONCLUSIONS

In conclusion, this nationwide longitudinal study underscores that TV may exacerbate the severity of OSA in moderate-severe OSA risk. Furthermore, air pollution can synergistically interact with TV to impact the OSA severity. The research findings expand our understanding of environmental risk factors affecting OSA severity and highlight the necessity of coordinated measures to address both temperature change and air pollution.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/envhealth.5c00045.

Descriptive statistics on TV (°C) as well as associations between main outcomes and TV over different lag periods, results of subgroup analyses and sensitivity analyses, and flowchart of participation inclusion in this study (PDF)

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Author Contributions

X.M., R.C., and Y.G. are joint corresponding authors and contributed equally to conceptualization, project administration, funding acquisition, resources, supervision, and writing-review and editing. A.L. and H.W. are joint first authors with equal contribution to data curation, formal analysis, investigation, writing-original draft, and writing-review and editing. Q.Z., X.Z., C.X., S.S., and H.K. contributed to software, methodology, validation, and writing-review and editing. A.L. and H.W. contributed equally to this work.

Notes

This study protocol received approval from the Central Medical Ethics Committee of the Chinese People's Liberation Army General Hospital (S2017-105-02) and all participants provided informed consent. The data used in this analysis were completely anonymous.

The authors declare no competing financial interest.

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